claude-windows / 6a82308f-69d6-43a8-9ea4-d8eaf497253d

Metadata

- Source: `claude-windows`
- Kind: `claude`
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\6a82308f-69d6-43a8-9ea4-d8eaf497253d\subagents\workflows\wf_3d536535-241\agent-a0c9bc13a56e85efe.jsonl`
- SHA-256: `0a5865522ac7537bf63090401c929f31841f3c0ad18bdeed308ebc0f7b427a83`
- Source modified: `2026-06-16T03:24:12+00:00`
- Imported at: `2026-07-05T16:48:27+00:00`
- project: `wf_3d536535-241`
- session_id: `6a82308f-69d6-43a8-9ea4-d8eaf497253d`

Transcript

1. user (2026-06-16T03:22:56.767Z)

Adversarial Claim Verifier (voter 3/3)

```
Be SKEPTICAL. Try to REFUTE this claim. ?2/3 refutations kill it.
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Research question

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Research question: **What concrete, recent (2022?2025) techniques and evaluation practices
should a local, commercially-licensed badminton rally-detector adopt to close the quality gap
with a strong cloud VLM (Gemini), given a tiny labeled set (~6 ? ~16 golden videos)?** Produce
an adversarially-verified, cited report organized around the four dimensions below. Prefer
specific named methods, papers (arXiv/venue), and reported numbers over generic advice; flag
anything you cannot verify.

CONTEXT (so you target the real gap, not generic ML):
- System: a local pipeline detects the shuttle per-frame (WASB / TrackNet, MIT-licensed) ? a
heuristic turns the trajectory into candidate rally windows ? today an optional Gemini "handoff"
confirms rallies. We just measured a fully-local (no-AI) run vs Gemini on 3 matches: local emits
far FEWER but MUCH LONGER windows (mean 65s vs 6.1s; one window spanned a whole 174s clip) ? it
does NOT miss activity, it FAILS TO SEGMENT. Root cause (code-confirmed): a frame is "active"
iff the shuttle is visible AND moving above a small per-second-normalized velocity threshold;
active frames within a 2 s gap are merged; there is NO max-window cap and NO inactivity/boundary
split. So "shuttle is moving" is conflated with "rally in progress," and dead-time + serves +
knock-ups bridge rallies into mega-windows. Shuttle tracking itself is excellent (96?99% per-
frame visibility).
- HARD CONSTRAINTS: weights/data/code must be MIT/Apache-2.0/BSD only (no viral/NC/custom
licenses, reject Gemma-3/Llama-4-MAU-cap); we may NEVER train owned weights on paid-LLM
(Gemini/OpenAI/Anthropic) outputs (terms/copyright), though paid LLMs may be used at inference;
minors excluded from any training pool; runs on a single local GPU.
- WHAT WE ALREADY HAVE (do NOT re-survey the basics ? go deeper/newer): the architecture
skeleton is known to be standard Temporal Action Localization/Segmentation (proposal+scorer) +
late/score-level fusion + stacked generalization; in-domain prior art (MonoTrack, ShuttleSet,
See et al. 2024/25 non-broadcast badminton rally start/end, Ghosh et al. point-segmentation,
tennis/TT play-break state machines) is already cited. We have a working eval harness: leave-
one-video-out (grouped by match), temporal-IoU F1 @0.5, F1@{0.1,0.25,0.5}, mAP@tIoU, segment-
count-ratio, bootstrap 95% CIs, paired exact sign-test, Platt/isotonic calibration + Brier/ECE,
a tiny numpy LogisticRegression scorer over per-window feature columns, an experiment
leaderboard, and an "eval?serve" guard (a cue measured +0.074 F1 on permissive proposal
windowing but ?0.008 on the coarser served windowing ? reverted). Available but default-OFF per-
window signals: net-crossing count (rally gate), 5 person/court features, 3 audio-hit features.

THE FOUR DIMENSIONS:

1) WINDOWING & BOUNDARIES (Temporal Action Localization/Segmentation): beyond proposal+score ?
what are the best *recent* methods for (a) turning a dense per-frame signal into well-segmented
```

actions, (b) explicitly controlling OVER-segmentation/merging, (c) boundary detection/regression and splitting merged segments? Cover e.g. MS-TCN/MS-TCN++, ASRF (action-boundary regression), BMN/BSN boundary-matching, ActionFormer/TriDet/TemporalMaxer (actionness + boundary heads), smoothing/boundary losses, and the standard metrics for over-segmentation (segmental F1@k, edit/Levenshtein score, mAP@tIoU). Which transfer to a 1-D shuttle-trajectory + auxiliary-cue setting on tiny data?

2) SHUTTLE-TRAJECTORY ? RALLY SIGNAL PROCESSING (racket-sports specific): how do published systems convert ball/shuttle tracking into rally/point segments and distinguish "in-play" from dead-time? Cover rally-state machines, serve/let detection, net-crossing & two-sided-occupancy cues, hit/contact (audio + kinematic) detection, and fps-invariant trajectory features. What features best separate "rally" from "shuttle merely moving"?

3) SMALL-DATA STRATEGIES: with ~6?16 labeled videos, what gives the most quality per label ? active learning (which videos/segments to label next; uncertainty/diversity/core-set), semi-/weakly-/point-supervised TAL (e.g. SF-Net, timestamp supervision), self-training/pseudo-labeling, and data-efficient TAL? Separately: the legally-clean ways to exploit a strong teacher (Gemini) at INFERENCE for weak/pseudo-labels or as an evaluation oracle WITHOUT its outputs becoming training data for owned weights ? what does the literature say about teacher-student / distillation boundaries and label-noise handling, and what governance distinctions matter?

4) EVALUATION METHODOLOGY & RIGOR at small n: trustworthy measurement with ~6?16 videos ? bootstrap/permutation tests, paired sign-tests, leave-one-group-out vs RLOOCV bias, calibration (reliability, ECE), the over-segmentation-blindness of plain tIoU, shadow/A-B evaluation, regression gates, and concrete pitfalls (label noise, annotator agreement/IAA, multiple-comparison/over-fitting traps, eval-vs-serve drift). What metrics + protocols should a small-data sports-segmentation leaderboard standardize on?

Deliverable: a structured, cited report with, per dimension, the 3?5 highest-leverage techniques that APPLY to our constrained setting (small data, local, license-clean, 1-D trajectory + cues + learned scorer), each with: what it is, why it fits our over-merge/small-data problem, the expected difficulty/payoff, and the citation. Call out which ideas are low-effort high-confidence wins vs research bets. Be explicit about anything unverified or license-incompatible.

Claim under review

"Frame-wise accuracy (Mean-over-Frames/MoF) does NOT capture over-segmentation ? it can score high even when a single action is fragmented into many discontinuous sub-segments ? whereas segment-based measures, the Edit score (Levenshtein-based) and F1@tau, are the appropriate metrics for detecting over-segmentation."

Source: https://arxiv.org/pdf/2210.10352 (primary)
Supporting quote: "MoF, as a per-frame calculation, does not capture segment quality; the score can be high even when the segments are fragmented. Dividing an action into many discontinuous sub-segments is referred to as over-segmentation. Segment-based measures like Edit-score [65] and F1-score [52] are instead more suitable measures for evaluating phenomenon like over-segmentation."

Checklist

1. Is the claim actually supported by the quote, or is it an overreach/misread?
2. WebSearch for contradicting evidence ? does any credible source dispute or heavily qualify this?
3. Is the source quality sufficient for the claim's strength? (extraordinary claims need primary sources)
4. Is the claim outdated? (check dates ? old claims about fast-moving fields are suspect)
5. Is this a marketing claim / press release / cherry-picked benchmark / forum speculation?

refuted=true if: unsupported by quote / contradicted / low-quality source for strong claim / outdated / marketing fluff.
refuted=false ONLY if: claim is well-supported, current, and source quality matches claim strength.
Default to refuted=true if uncertain.

Structured output only. Evidence MUST be specific.

2. assistant (2026-06-16T03:22:59.371Z)

I'll verify this claim about MoF (Mean-over-Frames) accuracy failing to capture over-segmentation versus segment-based metrics like Edit score and F1@tau.

3. assistant (2026-06-16T03:23:02.465Z)

The claim cites arXiv 2210.10352. Let me verify the source and the supporting quote, then search for any contradicting evidence.

4. user (2026-06-16T03:23:08.992Z)

I've reviewed the provided content, which appears to be a PDF document containing technical content. However, the material provided is primarily encoded PDF stream data and image content that doesn't contain readable text discussing the specific evaluation metrics you're asking about (MoF, over-segmentation, Edit score, and F1 score).

To properly answer your question about what the paper says regarding:
- Whether MoF captures over-segmentation
- Whether Edit-score and F1-score are more suitable alternatives

I would need access to the actual readable text content of the document. The current format doesn't allow me to extract and quote the relevant passages about these action segmentation evaluation metrics.

Could you provide the paper in a more accessible format (such as plain text or a clearer PDF extraction) so I can locate and quote the specific discussions about these metrics?

[Binary content (application/pdf, 1.8MB) also saved to C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\6a82308f-69d6-43a8-9ea4-d8eaf497253d\tool-results\webfetch-1781579999760-41alwg.pdf]

5. user (2026-06-16T03:23:12.310Z)

Web search results for query: "MoF frame-wise accuracy does not capture over-segmentation Edit score F1 action segmentation metric"

Links: [{"title":"Temporal Action Segmentation: An Analysis of Modern Techniques","url":"https://arxiv.org/pdf/2210.10352"},{"title":"GitHub - nus-cvml/awesome-temporal-action-segmentation: A curated list of awesome temporal action segmentation resources. · GitHub","url":"https://github.com/nus-cvml/awesome-temporal-action-segmentation"},{"title":"Temporal Action Segmentation: A Comprehensive Overview","url":"https://www.emergentmind.com/topics/temporal-action-segmentation"},{"title":"Alleviating Over-segmentation Errors by Detecting Action Boundaries","url":"https://arxiv.org/pdf/2007.06866"},{"title":"Video action segmentation by mixed temporal domain adaption","url":"https://image-ppubs.uspto.gov/dirsearch-public/print/downloadPdf/11138441"},{"title":"Improving action segmentation via explicit similarity measurement","url":"https://arxiv.org/html/2502.10713"},{"title":"Combining Boundary Supervision and Segment-Level Regularization for Fine-Grained Action Segmentation","url":"https://arxiv.org/html/2604.01859"},{"title":"Continuous Human Action Recognition for Human-Machine Interaction: A Review","url":"https://arxiv.org/pdf/2202.13096"},{"title":"Action Segmentation with Mixed Temporal Domain Adaptation","url":"https://arxiv.org/pdf/2104.07461"}]

Based on the search results, I found comprehensive information about this topic. Here's what the literature shows:

Key Points About MoF and Over-segmentation

MoF, as a per-frame calculation, does not capture segment quality; the score can be high even when the segments are fragmented. Dividing an action into many discontinuous sub-segments is referred to as over-segmentation.

Frame-wise accuracy is not sensitive to over-segmentation errors. More specifically, frame-wise accuracy is biased toward longer segments and underestimates over-segmentation errors.

Alternative Metrics for Evaluating Segmentation Quality

Segment-based measures like Edit-score and F1-score are more suitable measures for evaluating phenomena like over-segmentation.

Specifically:
- The Edit Score quantifies the similarity of two sequences based on the Levenshtein or edit distance and tallies the minimum number of insertions, deletions, and replacement operations required to convert one segment sequence into another.

- The segmental edit score penalizes over-segmentation by measuring the ordering of predicted action segments independent of slight temporal shifts.

These segment-based metrics provide a more accurate evaluation of action segmentation quality than frame-wise accuracy alone.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

6. assistant (2026-06-16T03:23:19.914Z)

The PDF binary wasn't parseable via WebFetch, but the search confirmed the source paper. Let me confirm the exact quote text and the paper's identity, then look for any contradicting/qualifying evidence.

7. user (2026-06-16T03:23:26.099Z)

Web search results for query: ""Temporal Action Segmentation" survey 2210.10352 MoF "does not capture" over-segmentation Edit-score F1-score"

Links: [{"title":"Temporal Action Segmentation: An Analysis of Modern ...","url":"https://arxiv.org/pdf/2210.10352"},{"title":"[2210.10352] Temporal Action Segmentation: An Analysis of Modern Techniques","url":"https://arxiv.org/abs/2210.10352"},{"title":"GitHub - nus-cvml/awesome-temporal-action-segmentation: A curated list of awesome temporal action segmentation resources. · GitHub","url":"https://github.com/nus-cvml/awesome-temporal-action-segmentation"},{"title":"Temporal Action Segmentation: An Analysis of Modern Techniques - ADS","url":"https://ui.adsabs.harvard.edu/abs/2022arXiv221010352D/abstract"},{"title":"awesome-temporal-action-segmentation/README.md at main · nus-cvml/awesome-temporal-action-segmentation","url":"https://github.com/nus-cvml/awesome-temporal-action-segmentation/blob/main/README.md"},{"title":"Temporal Action Segmentation: An Analysis of Modern Techniques","url":"https://www.researchgate.net/publication/374971079_Temporal_Action_Segmentation_An_Analysis_of_Modern_Techniques"},{"title":"Temporal Action Segmentation: An Analysis of Modern Technique","url":"https://www.researchgate.net/publication/364437795_Temporal_Action_Segmentation_An_Analysis_of_Modern_Technique"},{"title":"Temporal Action Segmentation: A Comprehensive Overview","url":"https://www.emergentmind.com/topics/temporal-action-segmentation"},{"title":"Hierarchical and Multimodal Data for Daily Activity Understanding","url":"https://arxiv.org/pdf/2504.17696"}]

Based on the search results, I found relevant information about your query regarding the Temporal Action Segmentation survey (arXiv:2210.10352).

Mean over frames (MoF) measures the proportion of correctly predicted frames but does not account for over-segmentation. This directly addresses your query about MoF not capturing over-segmentation.

The survey describes three main evaluation metrics: MoF estimates how accurate frame wise predictions are, Edit score tolerates small boundary shifts, as long as the sequence order is correct, and F1 score is the harmonic mean of the precision and recall of the detected action segments.

The frame-based measure (MoF) and the segment-based measures (Edit score and F1) differ, with the segment-based measures being more sensitive to over-segmentation issues that MoF does not capture. These metrics quantify both local classification accuracy and the coherence of temporal grouping, explicitly penalizing over-segmentation and boundary errors.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

8. user (2026-06-16T03:23:27.011Z)

Web search results for query: "frame-wise accuracy biased toward longer segments underestimates over-segmentation errors action segmentation"

Links: [{"title":"Combining Boundary Supervision and Segment-Level Regularization for Fine-Grained Action Segmentation","url":"https://arxiv.org/pdf/2604.01859"},{"title":"Video action segmentation by mixed temporal domain adaption","url":"https://image-ppubs.uspto.gov/dirsearch-public/print/downloadPdf/11138441"},{"title":"Action Segmentation with Mixed Temporal Domain Adaptation","url":"https://arxiv.org/pdf/2104.07461"},{"title":"Continuous Human Action Recognition for Human-Machine Interaction: A Review","url":"https://arxiv.org/pdf/2202.13096"},{"title":"Text-Derived Relational Graph-Enhanced Network for Skeleton-Based Action Segmentation","url":"https://arxiv.org/pdf/2503.15126"},{"title":"Unified Fully and Timestamp Supervised Temporal Action Segmentation via   Sequence to Sequence Translation","url":"https://arxiv.org/pdf/2209.00638"},{"title":"Bridge-Prompt: Towards Ordinal Action Understanding in Instructional Videos","url":"https://arxiv.org/pdf/2203.14104"},{"title":"Language-Assisted Human Part Motion Learning for Skeleton-Based Temporal   Action Segmentation","url":"https://arxiv.org/pdf/2410.06353"},{"title":"An Efficient Framework for Few-shot Skeleton-based Temporal Action Segmentation","url":"https://arxiv.org/pdf/2207.09925"},{"title":"Maximization and restoration: Action segmentation through dilation passing and temporal reconstruction - ScienceDirect","url":"https://www.sciencedirect.com/science/article/abs/pii/S003132032200245X"}]

Based on the web search results, I found several relevant sources discussing this topic in the context of action segmentation evaluation metrics.

Frame-wise accuracy measures the proportion of correctly classified frames, but it is biased toward longer segments and underestimates over-segmentation errors. This is a well-documented limitation in the action segmentation field.

While frame-wise accuracy is one of the most common evaluation metrics for action segmentation, it does not take into account the temporal dependencies of the prediction, causing large qualitative differences with similar frame-wise accuracy. In addition, long action classes have higher impact on this metric than shorter action classes, making this metric not able to reflect over-segmentation errors.

To address these limitations, researchers have adopted alternative metrics. Segmental F1 is emphasized, which better captures segmental quality and temporal consistency. Additionally, the segmental edit score penalizes over-segmentation by measuring the ordering of predicted action segments independent of slight temporal shifts.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

9. assistant (2026-06-16T03:23:36.150Z)

The claim is now well-corroborated. Let me note the key findings:

1. **Source identity confirmed**: arXiv 2210.10352 = "Temporal Action Segmentation: An Analysis of Modern Techniques" (Ding et al.), a peer-reviewed survey (IEEE TPAMI). The supporting quote matches the paper's text.

2. **Claim well-supported by quote**: The quote directly states MoF "does not capture segment

quality; the score can be high even when the segments are fragmented" and that "Segment-based measures like Edit-score and F1-score are instead more suitable measures for evaluating phenomenon like over-segmentation." The claim is a faithful paraphrase, not an overreach.

3. **Independently corroborated** by multiple other sources ? this is foundational, well-established consensus in the action segmentation literature, not a contested or extraordinary claim. The original Edit/F1@k metric proposals (Lea et al. ED-TCN, Lea et al. segmental scores) and ASRF paper (2007.06866) all reinforce it.

4. **Not outdated / not marketing**: This is a 2022 survey reflecting a decade of consensus; the principle has only been reinforced since (e.g., ASRF, TAS works through 2024-2025). It's academic, not a press release.

Let me do one quick adversarial check ? is there any nuance that would qualify the claim (e.g., that Edit/F1 also have their own blind spots, or that the claim is too strong)?

10. user (2026-06-16T03:23:47.340Z)

Web search results for query: "Edit score F1@k limitations also insensitive boundary action segmentation metric criticism temporal IoU"

Links: [{"title":"Action Segmentation with Mixed Temporal Domain Adaptation","url":"https://arxiv.org/pdf/2104.07461"},{"title":"Video action segmentation by mixed temporal domain adaption","url":"https://image-ppubs.uspto.gov/dirsearch-public/print/downloadPdf/11138441"},{"title":"GitHub - nus-cvml/awesome-temporal-action-segmentation: A curated list of awesome temporal action segmentation resources. · GitHub","url":"https://github.com/nus-cvml/awesome-temporal-action-segmentation"},{"title":"Gesture Recognition in Robotic Surgery: a Review","url":"https://arxiv.org/pdf/2102.00027"},{"title":"Temporal Convolutional Networks for Action Segmentation and Detection","url":"https://arxiv.org/pdf/1611.05267"},{"title":"Continuous Human Action Recognition for Human-Machine Interaction: A Review","url":"https://arxiv.org/pdf/2202.13096"},{"title":"Point-Supervised Skeleton-Based Human Action Segmentation","url":"https://arxiv.org/pdf/2603.06201"},{"title":"Temporal Action Segmentation: A Comprehensive Overview","url":"https://www.emergentmind.com/topics/temporal-action-segmentation"},{"title":"Alleviating Over-segmentation Errors by Detecting Action Boundaries","url":"https://arxiv.org/pdf/2007.06866"},{"title":"Spectral Scalpel: Amplifying Adjacent Action Discrepancy via Frequency-Selective Filtering for Skeleton-Based Action Segmentation","url":"https://arxiv.org/pdf/2603.24134"}]

Based on the search results, here's information about the Edit score, F1@k, and related temporal IoU metrics for action segmentation:

Overview of Metrics

The two most common segmental metrics designed to evaluate the temporal structure of action predictions are the edit score and the segmental F1 score with threshold k/100 (F1@k).

F1@k Metric Properties

The segmental F1 score has the following properties: (1) it penalizes over-segmentation errors, (2) it does not penalize for minor temporal shifts between the predictions and ground truth, which may have been caused by annotator variability, and (3) scores are dependent on the number actions and not on the duration of each action instance.

Limitations Addressed by These Metrics

These metrics address key limitations of frame-wise accuracy. Frame-wise accuracy does not take into account the temporal dependencies of the prediction, causing large qualitative differences with similar frame-wise accuracy, and long action classes have higher impact on this metric than shorter action classes, making this metric not able to reflect over-segmentation errors.

Insensitivity to Boundary Issues

The segmental edit score penalizes over-segmentation by measuring the ordering of predicted action segments independent of slight temporal shifts, and F1@k also penalizes over-segmentation errors while ignoring minor temporal shifts between the predictions and ground truth.

These metrics provide a more robust evaluation framework for action segmentation compared to simple frame-level accuracy.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

11. assistant (2026-06-16T03:23:57.723Z)

The adversarial check finds no contradiction ? it actually further reinforces the claim. The only minor nuance (Edit/F1@k themselves ignore minor temporal shifts and don't capture fine boundary precision) does not contradict the specific claim under review, which is narrowly about *over-segmentation*. On over-segmentation specifically, both metrics are designed to penalize it, exactly as claimed.

I attempted to refute this claim through five angles and failed on all of them:

1. **Quote support**: The claim is a faithful paraphrase of the supporting quote ? not an overreach. The quote explicitly says MoF "can be high even when the segments are fragmented" and names Edit-score and F1-score as "more suitable measures for evaluating phenomenon like over-segmentation."
2. **Contradicting evidence**: None found. Multiple independent sources (ASRF 2007.06866, ED-TCN/segmental-score originators Lea et al., MTDA, multiple 2022-2025 TAS papers) corroborate it.
3. **Source quality**: Strong. arXiv 2210.10352 = "Temporal Action Segmentation: An Analysis of Modern Techniques" (Ding et al.), a TPAMI-published survey ? a primary, authoritative reference. The claim is not extraordinary; it is field consensus.
4. **Outdated**: No. 2022 survey reinforced through 2025; principle dates to 2016 (Lea et al.) and has only strengthened.
5. **Marketing/cherry-pick**: No. Academic survey, methodological consensus, not a benchmark boast.

The only caveat (Edit/F1 have their own blind spots re: fine boundary localization, which is why tIoU-based mAP is added) does not weaken the specific over-segmentation claim ? it complements it.

12. user (2026-06-16T03:24:08.776Z)

Structured output provided successfully

13. assistant (2026-06-16T03:24:12.029Z)

Verification complete. The claim survived all adversarial refutation attempts ? **refuted=false** with high confidence. It is a faithful paraphrase of the primary source (arXiv 2210.10352, a TPAMI survey), independently corroborated by multiple action-segmentation papers, current, and not marketing. The only nuance (Edit/F1@k have their own boundary-localization blind spots) is orthogonal to the over-segmentation claim and does not weaken it.